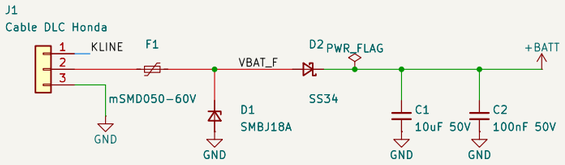
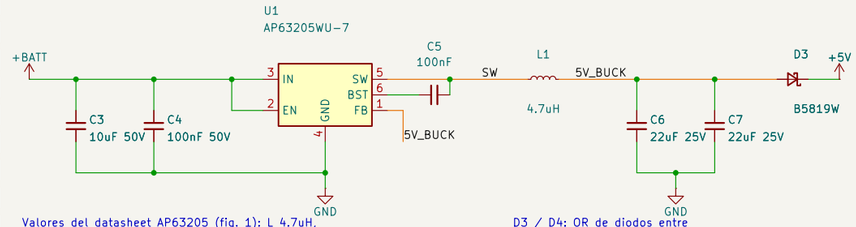
ENTRADA DLC HONDA + PROTECCION



Cable adaptador Honda 3p -> OBD2 cortado, soldado a J1:
1 = K-Line (OBD2 p7) 2 = +12V (p16) 3 = GND (p4/5)

PTC 0.5A + TVS 18V (descarga de carga)
+ Schottky en serie (polaridad invertida)

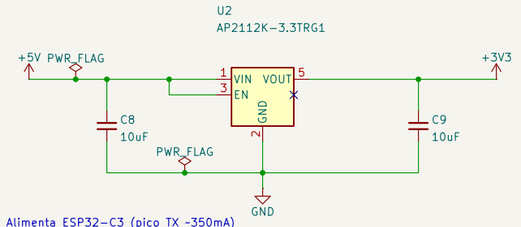
BUCK 12V -> 5V (AP63205, 2A)



Valores del datasheet AP63205 (fig. 1): L 4.7uH.
Cin 10uF, Cout 2x22uF, BST 100nF. EN atado a VIN.

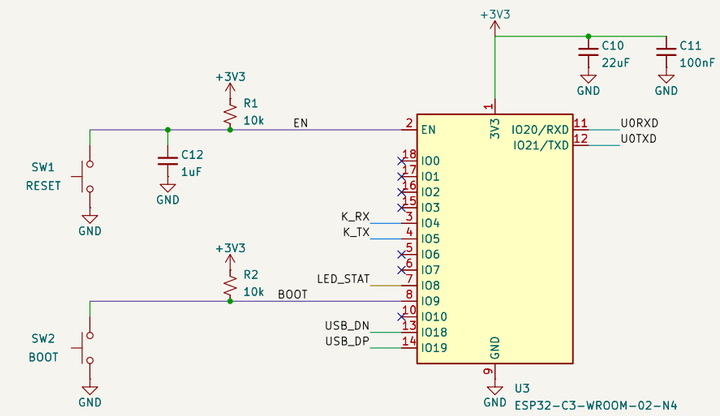
D3 / D4: OR de diodos entre
5V del buck y VBUS del USB

LDO 3.3V (AP2112K, 600mA)



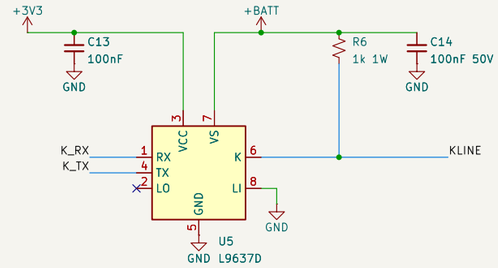
Alimenta ESP32-C3 (pico TX ~350mA)
y VCC del L9637D.

MCU ESP32-C3-WROOM-02 (BLE)



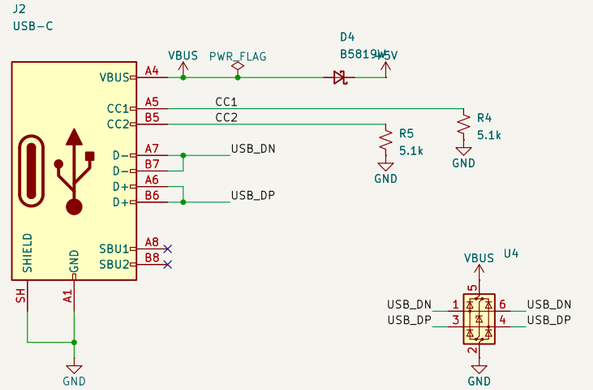
IO9 = BOOT (pull-up 10k + boton a GND). IO8 debe estar ALTO al arrancar (R3).
IO2 libre (strap con pull-up interno). USB nativo: IO18 = D-, IO19 = D+.
EN: RC 10k / 1uF (rampa del LDO < 1 ms).

TRANSCEPTOR K-LINE (L9637D, ISO 9141)



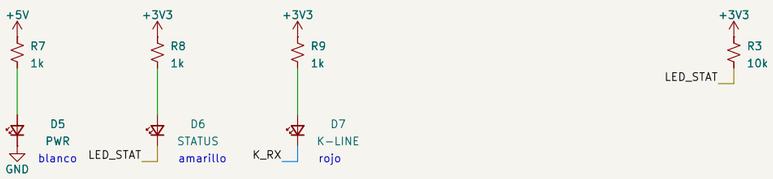
VCC = 3.3V: RX/TX con niveles del ESP32 (VCC 3-7V, VIH(TX) 2.5V).
Pull-up K 1k / 1W a VS (datasheet: RKO <= 5k).
LI a GND (comparador L sin uso, LO sin conectar).

USB-C (programacion / alimentacion)



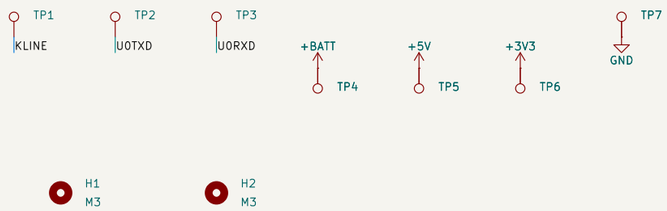
CC1/CC2 5.1k: sink USB 2.0. ESD en D+/D-/VBUS (U4).

LEDs INDICADORES



PWR: +5V presente. STATUS (IO8): activo en bajo: R3 asegura IO8 alto al arrancar.
K-LINE: parpadea con el trafico del bus (salida RX del L9637D), sin firmware. IO6/IO7 libres.

PUNTOS DE PRUEBA / MECANICO



Generado por hardware/scripts/gen_sch.py
honda-civic-scanner

Sheet: /
File: civic_scanner.kicad_sch

Title: Escaner Honda Civic K-Line (ESP32-C3 + L9637D)

Size: A3	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 10.0.8		Id: 1/1